1. An Explainable AI model uses an attention mechanism to determine the importance of four input features. For a particular query, the raw attention scores are:

$$S = [2, \ , 1, \ 0 \ 1]$$

The corresponding scalar value vectors are:

$$V = [8, \ , 5, \ 3 \ 6]$$

The attention weights are calculated using:

$$\alpha_i = \frac{e^{s_i}}{\sum_j e^{s_j}}$$

Use:

$$e^2 = 7.389, \ e^1 = 2.718, \ e^0 = 1$$

Answer the following:

a) Calculate the softmax attention weight for each input feature. [4 Marks]

b) Calculate the final attention output using

$$z = \sum_i \alpha_i V_i$$

[3 Marks]

c) Identify the most influential input feature and determine its percentage attention. [2 Marks]

d) Briefly interpret the result from an XAI perspective. [1 Mark]

Answer

Total exponential score:

$$= 7.389 + 2.718 + 1 + 2.718$$
$$= 13.825$$

Therefore,

$$\alpha_1 = \frac{7.389}{13.825} = 0.535$$
$$\alpha_2 = \frac{2.718}{13.825} = 0.197$$
$$\alpha_3 = \frac{1}{13.825} = 0.072$$
$$\alpha_4 = \frac{2.718}{13.825} = 0.197$$

Hence, $\alpha = [0.535, \ , 0.197, \ 0.072 \ 0.197]$

This follows the softmax-based attention calculation described in the material.

For the attention output:

$$z = (0.535)(8) + (0.197)(5) + (0.072)(3) + (0.197)(6)$$
$$= 4.280 + 0.985 + 0.216 + 1.182$$

$$z = 6.663$$

Feature 1 has the largest attention weight:

$$0.535 \times 100 = \boxed{53.5\%}$$

Final Answer: Attention weights = [0.535, 0.197, 0.072, 0.197], attention output = 6.663, and Feature 1 is the most influential with approximately 53.5% attention.

Feature Importance and Sensitivity Analysis [10 Marks]

2. A medical classification model has a baseline accuracy of 94%. To explain its prediction, permutation feature importance is applied to four features F_1, F_2, F_3, F_4.

After individually permuting the features, the following accuracies are obtained:

Feature permuted	Accuracy
F_1	0.72
F_2	0.88
F_3	0.91
F_4	0.80

Permutation importance is defined as:

$$I_j = \text{Baseline Accuracy} - \text{Permuted Accuracy}$$

For the same instance, the original model output is

$$f(x) = 0.75$$

When F_1 is increased by $\Delta = 0.05$, the output becomes 0.82. When F_4 is increased by the same amount, the output becomes 0.78.

Answer the following:

a) Calculate the permutation importance of all four features. [4 Marks]

b) Rank the four features from most important to least important. [2 Marks]

c) Calculate the finite-difference sensitivity of F_1 and F_4. [2 Marks]

$$S_i = \frac{| f(x + \Delta e_i) - f(x) |}{| \Delta |}$$

d) Which feature is more sensitive, and by what factor? [2 Marks]

Answer

The uploaded material defines permutation importance as the decrease in model performance after a feature is permuted.

Baseline:

$$A = 0.94$$

For F_1:

$$I_1 = 0.94 - 0.72 = \boxed{0.22}$$

For F_2:

$$I_2 = 0.94 - 0.88 = \boxed{0.06}$$

For F_3:

$$I_3 = 0.94 - 0.91 = \boxed{0.03}$$

For F_4:

$$I_4 = 0.94 - 0.80 = \boxed{0.14}$$

Therefore, $F_1 > F_4 > F_2 > F_3$

Sensitivity

For F_1:

$$S_1 = \frac{|\ 0.82 - 0.75\ |}{0.05}$$
$$= \frac{0.07}{0.05}$$
$$\boxed{S_1 = 1.4}$$

For F_4:

$$S_4 = \frac{|\ 0.78 - 0.75\ |}{0.05}$$
$$= \frac{0.03}{0.05}$$
$$\boxed{S_4 = 0.6}$$

The sensitivity formulation is consistent with the finite-difference method in the material.

Sensitivity ratio:

$$\frac{S_1}{S_4} = \frac{1.4}{0.6} = \boxed{2.33}$$

$F_1$is the most important feature according to permutation importance. It is also approximately 2.33 times more sensitive than F_4.

Question 3 — Grad-CAM and Explainability [10 Marks]

A CNN used for image classification generates the following two 2×2convolutional feature maps:

$$A^1 = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$
$$A^2 = \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$$

For a particular target class, the gradients with respect to these feature maps are:

$$G^1 = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$$
$$G^2 = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

Grad-CAM calculates each channel weight as

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

and constructs the heatmap as

$$L^c_{\text{Grad-CAM}} = ReLU\left(\sum_k \alpha_k^c A^k\right)$$

Answer the following:

a) Calculate $\alpha_1$and α_2. [3 Marks]

b) Compute $\alpha_1 A^1$and $\alpha_2 A^2$. [2 Marks]

c) Construct the final 2×2Grad-CAM heatmap. [3 Marks]

d) Identify the image location with the strongest class evidence and interpret it. [2 Marks]

Answer

The Grad-CAM procedure in the material first spatially averages the target-class gradients to obtain a weight for each channel, then computes the weighted feature-map sum and applies ReLU.

There are four spatial positions, so

$$Z = 4$$

For feature map 1:

$$\alpha_1 = \frac{2 + 1 + 1 + 0}{4}$$
$$= \frac{4}{4}$$
$$\boxed{\alpha_1 = 1.0}$$

For feature map 2:

$$\alpha_2 = \frac{0+1+0+1}{4}$$
$$= \frac{2}{4}$$
$$\boxed{\alpha_2 = 0.5}$$

Now,

$$\alpha_1 A^1 = 1.0 \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

and

$$\alpha_2 A^2 = 0.5 \begin{bmatrix} 1 & 2 \\ 2 & 0 \end{bmatrix}$$
$$= \begin{bmatrix} 0.5 & 1 \\ 1 & 0 \end{bmatrix}$$

Add the weighted feature maps:

$$L = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 0.5 & 1 \\ 1 & 0 \end{bmatrix}$$
$$= \begin{bmatrix} 2.5 & 2 \\ 1 & 3 \end{bmatrix}$$

All values are positive, so ReLU does not alter them: $L_{\text{Grad-CAM}} = \begin{bmatrix} 2.5 & 2.0 \\ 1.0 & 3.0 \end{bmatrix}$

This is the same heatmap-construction procedure illustrated in the source material.

The maximum activation is:

$$\max(L) = 3.0$$

at the bottom-right location.

Therefore, the bottom-right spatial region provides the strongest evidence for the target class.

SHAP Question and Answer

Question

A financial institution uses an Explainable Artificial Intelligence (XAI) system based on SHAP (SHapley Additive exPlanations) to explain the output of a machine-learning model that predicts the probability of loan approval for a particular applicant.

For a specific applicant, the model has a baseline/reference value:

$E[f(x)] = 0.40$

The SHAP values of five input features are:

$\varphi = [0.15, -0.10, 0.20, -0.05, 0.12]$

where the features represent:
- F_1: Annual Income
- F_2: Existing Debt
- F_3: Credit Score
- F_4: Loan Amount
- F_5: Employment Stability

Use the SHAP additive relationship:

$f(x) = E[f(x)] + \Sigma(i=1 \text{ to } 5)\ \varphi_i$

and consider the absolute SHAP value $|\varphi_i|$ for determining feature importance.

Calculate the following:
1. Calculate the final model prediction for the applicant.
2. Determine the absolute SHAP value $|\varphi_i|$ for each feature.
3. Rank the five features according to their importance.
4. Identify the most influential feature based on its absolute SHAP value.
5. Classify the features into positive and negative contributors to the prediction.
6. Calculate the total positive and negative contributions.
7. Explain how the SHAP values move the model prediction from the baseline value to the final model output.
8. Present the prediction movement step-by-step from the baseline through all five feature contributions.

Answer

Given:

Baseline value: $E[f(x)] = 0.40$

SHAP values: $\varphi = [0.15, -0.10, 0.20, -0.05, 0.12]$

1. Final Model Prediction

Using the SHAP additive relationship:

$f(x) = E[f(x)] + \Sigma\varphi_i$

$\Sigma\varphi_i = 0.15 - 0.10 + 0.20 - 0.05 + 0.12 = 0.32$

Therefore:

$f(x) = 0.40 + 0.32 = 0.72$

Final model prediction = 0.72.

2. Absolute SHAP Values

Feature	SHAP Value	Absolute SHAP Value
Annual Income (F_1)	+0.15	0.15
Existing Debt (F_2)	−0.10	0.10
Credit Score (F_3)	+0.20	0.20
Loan Amount (F_4)	−0.05	0.05
Employment Stability (F_5)	+0.12	0.12

3. Feature Importance Ranking

Ranking based on absolute SHAP value:

F_3 (Credit Score) > F_1 (Annual Income) > F_5 (Employment Stability) > F_2 (Existing Debt) > F_4 (Loan Amount)

Rank	Feature	$\lvert\varphi_i\rvert$
1	Credit Score (F_3)	0.20
2	Annual Income (F_1)	0.15
3	Employment Stability (F_5)	0.12
4	Existing Debt (F_2)	0.10
5	Loan Amount (F_4)	0.05

4. Most Influential Feature

Credit Score (F_3) is the most influential feature because it has the largest absolute SHAP value:

$|\varphi_3| = |0.20| = 0.20$

It has a positive contribution and increases the prediction by 0.20.

5. Positive and Negative Contributions

Positive contributors:

• Annual Income (F_1): +0.15
• Credit Score (F_3): +0.20
• Employment Stability (F_5): +0.12

Total positive contribution:

0.15 + 0.20 + 0.12 = +0.47

Negative contributors:

• Existing Debt (F_2): −0.10
• Loan Amount (F_4): −0.05

Total negative contribution:

−0.10 − 0.05 = −0.15

Net contribution = +0.47 − 0.15 = +0.32

6. Step-by-Step Movement from Baseline to Final Prediction

Baseline = 0.40

After F_1 (+0.15): 0.40 + 0.15 = 0.55

After F_2 (−0.10): 0.55 − 0.10 = 0.45

After F_3 (+0.20): 0.45 + 0.20 = 0.65

After F_4 (−0.05): 0.65 − 0.05 = 0.60

After F_5 (+0.12): 0.60 + 0.12 = 0.72

7. Final Interpretation

The model prediction increases from the baseline value of 0.40 to the final value of 0.72. The positive contributions from Credit Score, Annual Income, and Employment Stability increase the prediction, while Existing Debt and Loan Amount decrease it. Credit Score is the most influential feature because it has the largest absolute SHAP value of 0.20.